

Solving for the toponium Green's function with the method of Jeżabek, Kühn and Teubner

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Abstract

Artemiy's funky attempt at toponium. Near threshold, everything about a $t\bar{t}$ pair that feels the QCD potential is contained in one complex function, the S-wave Green's function $G(E, p)$. This note shows how I compute it from scratch by following [JKT] equation by equation: the potential of their Sec. 2, the regularisation and numerical solution of their Sec. 3, the checks that the solver is right, and what it gives when used to re-weight MadGraph events. Fuks et al. appear only as the point of comparison: they publish tables of G , but not how they made them.

References (PDFs in `reference/`)

- [JKT] M. Jeżabek, J.H. Kühn, T. Teubner, Z. Phys. C 56 (1992) 653 (`BF01474740.pdf`). *The method*.
- [Fuks] B. Fuks, K. Hagiwara, K. Ma, Y.-J. Zheng, arXiv:2411.18962v2 (`2411.18962v2.pdf`). *The comparison*.

Equation numbers always refer to the paper named in front of them: [JKT] (21) is Eq. (21) of JKT. The code is one library, `toponium.py`; a “ $\rightarrow$ name” at the end of a paragraph names the class or method that implements it.

1 The problem

A $t\bar{t}$ pair produced at a point with energy $E = W - 2m_t$ (W the pair's invariant mass) and relative momentum p propagates as

$$G(\vec{p}, E) = \int d^3r e^{i\vec{p}\cdot\vec{r}} G(\vec{r}, 0; E) = \left\langle \vec{p} \left| \frac{1}{\mathcal{E} - H} \right| \vec{r} = 0 \right\rangle,$$
$$H = \frac{p^2}{m_t} + V(r), \quad \mathcal{E} = E + i\Gamma_t \quad ([\text{JKT}] (2)-(4)).$$

The width enters through $E \rightarrow \mathcal{E}$: the tops decay before the pair can form a stable bound state, so the toponium “resonances” are smeared into a single bump. Without the potential,

$$G_0(p) = \frac{1}{\mathcal{E} - p^2/m_t} \quad ([\text{JKT}] (6)),$$

and the potential's whole effect is in the ratio G/G_0 . In particular, re-weighting each MadGraph $t\bar{t}$ event by $|G(E, p^*)/G_0(E, p^*)|^2$ ([Fuks] (15)) puts toponium into the event sample.

So the task is: *compute $G(E, p)$ for the QCD potential*. [JKT] give both the potential (their Sec. 2) and a numerical method (their Sec. 3), and I follow both as printed. Where they leave a choice open, I say so and list it in Sec. 8.

Contrast with Fuks et al. [Fuks] use the same re-weighting but do not say how they computed G ; they publish it as a table (swData) and cite [JKT]. My G is computed independently of those tables. Comparing the two turned out to be a test of the tables rather than of my solver (Sec. 6).

Conventions: $m_t = 173$ GeV, $\Gamma_t = 1.49$ GeV, $C_F = 4/3$, $n_F = 5$, $\alpha_s(m_Z) = 0.12$. G has the sign of [JKT] (6). $\rightarrow$ `Constants`, `Constants.GO`

2 The method at a glance

[JKT]	what it is	in toponium.py
(14)	two-loop running coupling $\alpha_s(Q^2)$	<code>JKTPotential.alpha_2loop,</code> <code>lambda_msbar_2loop</code>
(13)	perturbative potential $\tilde{V}_{\text{pert}}(Q^2)$	<code>JKTPotential.V_pert</code>
(15)	full potential: (13) above 5 GeV, Richardson-like below	<code>V_ir, continuity_constant_C,</code> <code>V_jkt</code>
(19)–(20)	cut at small momentum, and its normalisation	<code>V_cut, V_cut_position,</code> <code>energy_shift</code>
(16)–(17)	$G = G_0 K$ and the integral equation for K	<code>LSSolver.G</code>
(18)	S-wave kernel $\hat{V}(p, q)$	<code>LSSolver.kernel</code>
(21)–(22)	subtracted equation and $\mathcal{A}(p)$	<code>LSSolver.script_A, solve_K</code>
(23)	Gauss–Legendre grid	<code>LSSolver.jkt_grid</code>

Algorithm. Input: $\alpha_s(m_Z)$, m_t , Γ_t .		
1. Build the potential: $\Lambda_{\overline{\text{MS}}}$ from (14) at m_Z , then (13), (15), the cut (19).		Sec. 3
2. Fix the cut's δ^3 term by (20); it becomes a constant energy shift.		Sec. 3
3. Tabulate $F(s) = \int_0^s V dk^2$, which gives the kernel (18).		Step 2
4. Nodes and weights q_j, w_j (23); matrix $W_{ij} = w_j \hat{V}(q_i, q_j)$, zero diagonal.		Steps 2, 5
5. For each E : $g_j = G_0(q_j; E - \text{shift})$, $\mathcal{A}(q_i)$ (22), solve (21) for K_i .		Steps 3, 4, 6
6. For each output p : $G(p) = G_0(p) K(p)$ (16), K splined between nodes.		Step 7

3 The potential (JKT Sec. 2 and the cut of Sec. 3)

Everything here is in momentum space: $\tilde{V}(Q^2)$, with Q the momentum transfer. $\rightarrow$ `JKTPotential`

The running coupling, [JKT] (14).

$$\alpha_{\overline{\text{MS}}}(Q^2) = \frac{4\pi}{b_0 \ln(Q^2/\Lambda_{\overline{\text{MS}}}^2) + \frac{b_1}{b_0} \ln \ln(Q^2/\Lambda_{\overline{\text{MS}}}^2)}, \quad b_0 = 11 - \frac{2}{3}n_F, \quad b_1 = 102 - \frac{38}{3}n_F.$$

JKT quote α_s directly; I quote $\alpha_s(m_Z) = 0.12$, the value [Fuks] use, and *solve* (14) for Λ : with $x = \ln(m_Z^2/\Lambda^2)$, (14) at $Q = m_Z$ reads

$$b_0 x + \frac{b_1}{b_0} \ln x = \frac{4\pi}{\alpha_s(m_Z)},$$

a single equation in x , solved by root-finding (Brent's method). This is the only equation solved in building the potential; everything after it is evaluated directly. For $\alpha_s(m_Z) = 0.12$, $n_F = 5$: $\Lambda_{\overline{\text{MS}}}^{(5)} = 223$ MeV, and (14) then gives $\alpha_s = 0.221, 0.183, 0.157, 0.150$ at $Q = 5, 10, 20, 25$ GeV. $\rightarrow$ `beta_coefficients`, `lambda_msbar_2loop`, `alpha_2loop`

The perturbative potential, [JKT] (13).

$$\tilde{V}_{\text{pert}}(Q^2) = -\frac{16\pi}{3} \frac{\alpha_{\overline{\text{MS}}}(Q^2)}{Q^2} \left[1 + \left(\frac{31}{3} - \frac{10}{9} n_F \right) \frac{\alpha_{\overline{\text{MS}}}(Q^2)}{4\pi} \right].$$

This is one-gluon exchange, $-4\pi C_F \alpha / Q^2$ with $4\pi C_F = 16\pi/3$, with two improvements: the coupling runs with the momentum transfer, $\mu^2 = Q^2$, by (14), and the square bracket is the one-loop correction to the potential. Nothing is solved: for each Q^2 , compute $\alpha_{\overline{\text{MS}}}(Q^2)$ from (14), put it into (13). (With α held fixed this is the potential of [Fuks] (12); without the bracket and with fixed α it is plain Coulomb, the test case of Sec. 5.1.) → `V_pert`

Joining to the long-distance potential, [JKT] (15). (13) is valid only at large Q ; at small Q the coupling blows up at $Q = \Lambda$. Below $|\vec{p}| = 5$ GeV JKT switch to a Richardson-like form that also produces the linear confining potential:

$$\tilde{V}(\vec{p}) = \begin{cases} \tilde{V}_{\text{pert}}(p^2, n_F = 5), & |\vec{p}| > 5 \text{ GeV}, \\ -\frac{4}{3} \frac{48\pi^2}{27} \frac{1}{p^2} \left(\frac{1}{\ln(1 + p^2/\Lambda_R^2)} + C \right), & |\vec{p}| < 5 \text{ GeV}, \end{cases} \quad \Lambda_R = 400 \text{ MeV}.$$

C is fixed by continuity at 5 GeV: setting the two lines equal at $p^2 = 25 \text{ GeV}^2$ and solving for C (one linear equation) gives $C = -0.0262$. At small p , $1/\ln(1 + p^2/\Lambda_R^2) \approx \Lambda_R^2/p^2$, so $\tilde{V} \approx -\kappa \Lambda_R^2/p^4$ with $\kappa = \frac{4}{3} \cdot \frac{48\pi^2}{27}$: the Fourier transform of a potential rising linearly with r . → `V_ir`, `continuity_constant_C`, `V_jkt`

The cut, [JKT] (19). The $1/p^4$ behaviour is a problem: its Fourier integral diverges at $p \rightarrow 0$, and the kernel (18) built from it is singular as $(p - q)^{-2}$ at $q = p$. JKT argue that t and $\bar{t}$ decay before they reach the confining distances, so G cannot depend on how the potential behaves at very small momentum, and they replace it there:

$$\tilde{V}_{\text{cut}}(\vec{p}; q_{\text{cut}}) = \begin{cases} \tilde{V}(\vec{p}), & |\vec{p}| > q_{\text{cut}}, \\ -\frac{C_1}{p^2 + q_{\text{cut}}^2} + \tilde{V}_0(q_{\text{cut}}) \delta^3(\vec{p}), & |\vec{p}| < q_{\text{cut}}, \end{cases} \quad q_{\text{cut}} = 50 \text{ MeV}.$$

C_1 comes from continuity at $p = q_{\text{cut}}$: $-C_1/(2q_{\text{cut}}^2) = \tilde{V}(q_{\text{cut}}^2)$.

Normalising the cut, [JKT] (20). Cutting changes $V(r)$ by a constant. JKT fix $\tilde{V}_0$ by demanding that the position-space potential

$$V_{\text{cut}}(r; q_{\text{cut}}) = \int \frac{d^3 q}{(2\pi)^3} e^{-i\vec{q}\cdot\vec{r}} \tilde{V}_{\text{cut}}(\vec{q}; q_{\text{cut}}) = \frac{1}{2\pi^2 r} \int_0^\infty dq q \sin(qr) \tilde{V}_{\text{cut}}(q^2) + \frac{\tilde{V}_0}{(2\pi)^3}$$

satisfy $V_{\text{cut}}(r = 1 \text{ GeV}^{-1}) = -\frac{1}{4} \text{ GeV}$. The δ^3 term is just the constant $\tilde{V}_0/(2\pi)^3$ in $V(r)$, i.e. a constant added to H , so in the Green's function it is the same as $E \rightarrow E - \tilde{V}_0/(2\pi)^3$. I therefore solve with the smooth part of (19) and evaluate G_0 at the shifted energy. For $\alpha_s(m_Z) = 0.12$ the shift is $+5.678 \text{ GeV}$. It is large because it cancels the constant $\sim \kappa \Lambda_R^2/(2\pi^2 q_{\text{cut}})$ that the cut adds to $V(r)$; it depends on q_{cut} , but G does not (Sec. 5). → `V_cut`, `V_cut_position`, `energy_shift`

4 Solving the Lippmann–Schwinger equation (JKT Sec. 3)

Step 1: the equation, [JKT] (5), (16)–(17). The Green’s function obeys

$$G(\vec{p}) = G_0(p) + G_0(p) \int \frac{d^3 q}{(2\pi)^3} \tilde{V}(\vec{p} - \vec{q}) G(\vec{q}) \quad ([\text{JKT}] (5)).$$

For the S-wave JKT write $G = G_0 K$ (16), and the vertex function K obeys

$$K(p) = 1 + \int_0^\infty dq \hat{V}(p, q) G_0(q) K(q) \quad ([\text{JKT}] (17)).$$

K is smooth in p : the sharp structure of G (the on-shell peak at $p^2 = m_t E$ for $E > 0$) sits in G_0 , which is known exactly.

Step 2: the S-wave kernel, [JKT] (18). The angular integral of $\tilde{V}(|\vec{p} - \vec{q}|)$ gives

$$\begin{aligned} \hat{V}(p, q) &= \frac{1}{4\pi^2} \frac{q}{p} \int_{|p-q|}^{p+q} du u \tilde{V}(u) = \frac{q}{8\pi^2 p} \left[F((p+q)^2) - F((p-q)^2) \right], \\ F(s) &= \int_0^s dk^2 \tilde{V}(k^2). \end{aligned}$$

F is computed once per potential (10-point Gauss–Legendre on 6000 cells in $\ln s$, with the kinks of $\tilde{V}$ at q_{cut} and 5 GeV on cell edges) and splined. This is the only place the potential of Sec. 3 enters the solver. For pure Coulomb, $\tilde{V} = -B/k^2$, the integral is elementary: $\hat{V} = -\frac{Bq}{8\pi^2 p} \ln \frac{(p+q)^2}{(p-q)^2}$.
→ `LSSolver.kernel`

Step 3: the peak at $q = p$, and JKT’s subtraction, [JKT] (21)–(22). $\hat{V}(p, q)$ is large at $q = p$ (small momentum transfer). With the cut it is finite but peaked on the scale $q_{\text{cut}} = 50$ MeV; for Coulomb it has a logarithmic singularity. The grid nodes (Step 5) are ~ 0.1 – 0.25 GeV apart, so a plain Gauss–Legendre sum misses the peak (Fig. 1, left).

Why not simply put more nodes there? Because “there” is everywhere: the same nodes q_j serve every row $p = q_i$ of the linear system (Step 6), and the peak of row i sits at $q = q_i$, on a node of its own. Refining the grid around one p does nothing for the others; resolving a 50 MeV peak at every node would take a grid ~ 100 times finer, $N \sim 10^4$. JKT instead add and subtract $K(p)$ inside the integral of (17):

$$[1 - \mathcal{A}(p)] K(p) + \int_0^\infty dq \hat{V}(p, q) G_0(q) [K(p) - K(q)] = 1, \quad ([\text{JKT}] (21))$$

$$\mathcal{A}(p) = \int_0^\infty dq \hat{V}(p, q) G_0(q). \quad ([\text{JKT}] (22))$$

This is an identity, not an approximation. It moves the peak into $\mathcal{A}(p)$, which contains no unknown: in JKT’s words, “a known function which can be evaluated by a numerical integration”. In the remaining integral the peak is multiplied by $K(p) - K(q) \approx K'(p)(p - q)$, which vanishes at $q = p$. What is left is ~ 1000 times smaller and odd about $q = p$, so the contributions of the nodes on either side largely cancel (Fig. 1, right). The peak is then integrated *once per p*, in $\mathcal{A}(p)$, where a fine rule built around that one p is cheap (Step 4, Fig. 1, middle).

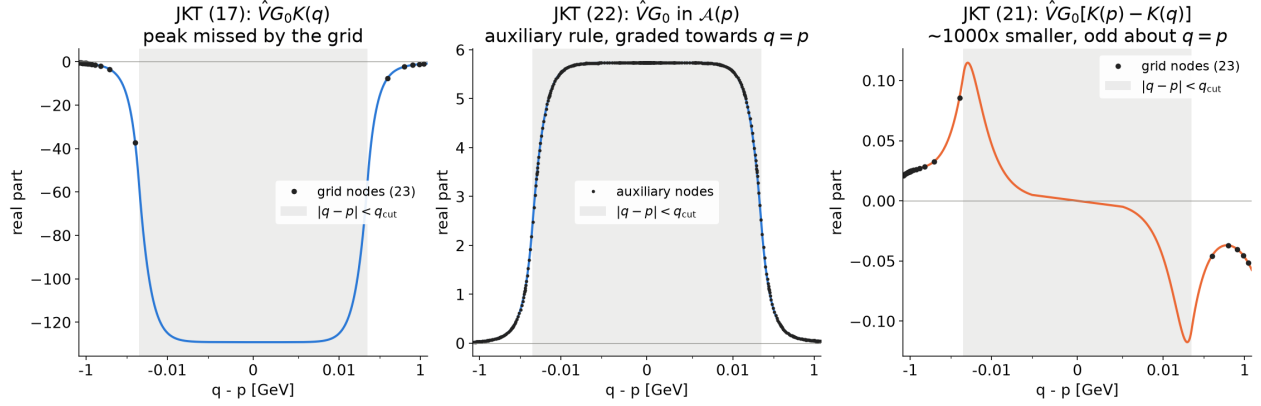


Figure 1: JKT potential ($\alpha_s(m_Z) = 0.12$, $q_{\text{cut}} = 50$ MeV), $p = 5$ GeV, $E = -2$ GeV, against $q - p$ on a logarithmic scale (linear for $|q - p| < 1$ MeV) so that the peak region, $|q - p| \lesssim q_{\text{cut}}$ (shaded), is spread out. Left: the integrand of [JKT] (17) with the grid nodes of [JKT] (23): the peak lies between two nodes. Middle: the integrand of [JKT] (22), on the auxiliary rule of $\mathcal{A}(p)$, whose panels are graded geometrically towards $q = p$ and so are evenly spaced on this axis. Right: the integrand of [JKT] (21), $\sim 1000\times$ smaller and odd about $q = p$, which the grid nodes integrate accurately.

Step 4: computing $\mathcal{A}(p)$, [JKT] (22). JKT do not say how. I use a fine auxiliary Gauss–Legendre rule for each p : 16-point panels with edges graded geometrically towards $q = p$ ($p(1 \pm 2^{-k})$, $k \leq 24$, i.e. evenly spaced in $\ln |q - p|$; Fig. 1, middle) to resolve the peak, further edges at $p(1 + 2^k)$ for its fall-off, every 2 GeV up to 200 GeV for the on-shell peak of G_0 , and a $t = 1/q$ tail. $\hat{V}$ on these points is stored once, so for each energy $\mathcal{A}(p)$ is one dot product with G_0 . Against adaptive quadrature the relative error is 2×10^{-10} (Coulomb) and 6×10^{-6} (JKT, limited by the spline of F).
 $\rightarrow \text{LSSolver.aux_rule, script_A}$

Step 5: the grid, [JKT] (23). Every $\int_0^\infty dq$ in (21) becomes a Gauss–Legendre sum $\sum_j w_j(\cdot)(q_j)$. JKT use three subintervals, $(0, a_1)$, (a_1, a_2) and $(0, 1/a_2)$ in $t = 1/q$, with $a_1 \sim \sqrt{m_t \Gamma_t}$ and $a_2 \sim m_t$. I split them once more, at 4 and 60 GeV: edges 0, 4, 16, 60, m_t plus the $t = 1/q$ tail, with $n = 48$ points each ($N = 240$; JKT used $N = 100\text{--}400$). The reason: above threshold at narrow width ($m_t = 120$, $\Gamma_t = 0.3$ GeV, JKT’s Fig. 4a at $E = 2$), the on-shell peak of G_0 at $p_0 = \sqrt{m_t E}$ is only $m_t \Gamma_t / 2p_0 \approx 1.2$ GeV wide. On JKT’s wide middle panel it falls between the nodes, and the peak of $|pG|^2$ goes $9272 \rightarrow 7680 \rightarrow 7219$ for $n = 48, 96, 160$ without settling. With the extra breaks it gives 7065, 7044, 7036 (JKT: ~ 7050).
 $\rightarrow \text{LSSolver.jkt_grid}$

Step 6: one linear system per energy. Imposing (21) at the nodes $p = q_i$ gives, with $g_j = G_0(q_j)$ and $W_{ij} = w_j \hat{V}(q_i, q_j)$,

$$\left[1 - \mathcal{A}(q_i) + \sum_{j \neq i} W_{ij} g_j\right] K_i - \sum_{j \neq i} W_{ij} g_j K_j = 1.$$

The $j = i$ term drops out exactly, because it multiplies $K_i - K_i = 0$. W does not depend on E and is built once. Each energy costs $\mathcal{A}$ (Step 4) and one dense $N \times N$ solve. $\rightarrow \text{LSSolver.solve_K}$

Step 7: G at any p , [JKT] (16). $G(p) = G_0(p)K(p)$, with K between the nodes from a cubic spline in $\ln q$ through the $K(q_i)$. Because K is smooth and G_0 is exact, this keeps the sharp structure

of G . (I also tried evaluating (21) itself at off-grid p . That is 10–40 times less accurate, because at a general p the nodes are not placed symmetrically around the peak of Step 3.) $\rightarrow$ `LSSolver.G`

5 Is the solver right?

5.1 Ground truth: the exact Coulomb solution

For a fixed-coupling Coulomb potential, $V = -C_F\alpha/r$, the equation can be solved on paper (JKT mention this below their (12)). The solution is a Whittaker function,

$$G(r) = -\frac{m_t}{4\pi r} \Gamma(1 - \lambda) W_{\lambda, 1/2}(2\kappa r), \quad \kappa = \sqrt{-m_t \mathcal{E}}, \quad \lambda = \frac{C_F \alpha m_t}{2\kappa},$$

and its Laplace representation plus the S-wave Fourier transform give a one-dimensional integral,

$$\tilde{G}(p) = -4m_t \kappa^2 \int_0^\infty dt \frac{t^{-\lambda}(1+t)^\lambda(1+2t)}{(\kappa^2(1+2t)^2 + p^2)^2},$$

which is G_0 at $\lambda \rightarrow 0$. It converges at $t \rightarrow 0$ only for $\text{Re } \lambda < 1$ ($\lambda = 1$ is the 1S pole), so it is continued by subtracting $f(0)$ on $[0, 1]$, valid for $\text{Re } \lambda < 2$; since $|\lambda| \leq 1.08$ for $\alpha = 0.15$, one subtraction suffices. No numerical solver is involved. $\rightarrow$ `CoulombExact`

5.2 Accuracy and q_{cut} independence (Stage 0)

On $E \in [-6, 4]$ GeV, $p \in [0.5, 29.5]$ GeV, $m_t = 173$, $\Gamma_t = 1.49$ GeV. The solver with a Coulomb potential is compared with the exact solution; with the JKT potential, against itself at $n = 96$:

n	N	Coulomb $\alpha = 0.15$ vs exact		JKT $\alpha_s(m_Z) = 0.12$ vs $n = 96$	
		median	max	median	max
16	80	5.3×10^{-5}	1.1×10^{-2}	6.5×10^{-4}	8.9×10^{-2}
24	120	1.4×10^{-5}	2.2×10^{-3}	2.4×10^{-4}	2.1×10^{-2}
32	160	5.9×10^{-6}	7.0×10^{-4}	1.2×10^{-4}	8.9×10^{-3}
48	240	1.8×10^{-6}	2.0×10^{-4}	5.4×10^{-5}	3.2×10^{-3}
64	320	7.5×10^{-7}	8.0×10^{-5}	2.5×10^{-5}	1.4×10^{-3}

The maxima sit above threshold on the on-shell ridge; for $E \leq 0$ they are 1.1×10^{-5} (Coulomb) and 5.8×10^{-4} (JKT) at $n = 48$. JKT quote “better than 1%, typically 10^{-3} ”.

q_{cut} independence (JKT: “ K is independent from q_{cut} for $q_{\text{cut}} < 200$ MeV”). Relative to $q_{\text{cut}} = 50$ MeV, G changes by at most 4.9×10^{-3} at 200 MeV, 1.6×10^{-3} at 100 MeV, 1.0×10^{-3} at 20 MeV and 1.3×10^{-3} at 10 MeV, while the energy shift of (20) ranges from 1.6 to 27 GeV. The cut does what JKT claim: it regularises without changing the physics.

5.3 Reproducing JKT’s own results (Stage 1)

Since this is JKT’s method, it must reproduce their figures (Fig. 2). I use $\alpha_s = 0.125$ as JKT do, and Γ_t from [JKT] (8)–(9): 0.299, 0.807 and 1.563 GeV for $m_t = 120, 150$ and 180 (JKT’s text rounds to 0.3/0.8/1.5). At $m_t = 120$ the 1S peak of R is at -2.32 GeV (JKT: -2.3) with height 3.80 (JKT: 3.78). The 1S–2S minimum, the 2S bump, $R(0)$ and $R(4)$ agree to 0.2–2%. The peaks of $|pG|^2$ in their Fig. 4a–c agree to 0–3% with the values read off by eye, and $-G(p \rightarrow 0)$ in their Fig. 2 to 0.4–3%. The exception is $\text{Im}(-G)(p \rightarrow 0)$ at $E = 0$ (Fig. 3): -2.98 against their -2.7 , i.e.

10%, on a small value on a ± 8 axis. The optical theorem [JKT] (12), $\int \frac{d^3p}{(2\pi)^3} \Gamma_t |G|^2 = -\text{Im } G(0, 0)$, holds to 4×10^{-5} for $E \in [-5, 5]$ GeV; it is not built into the solver, so it independently checks the solution.

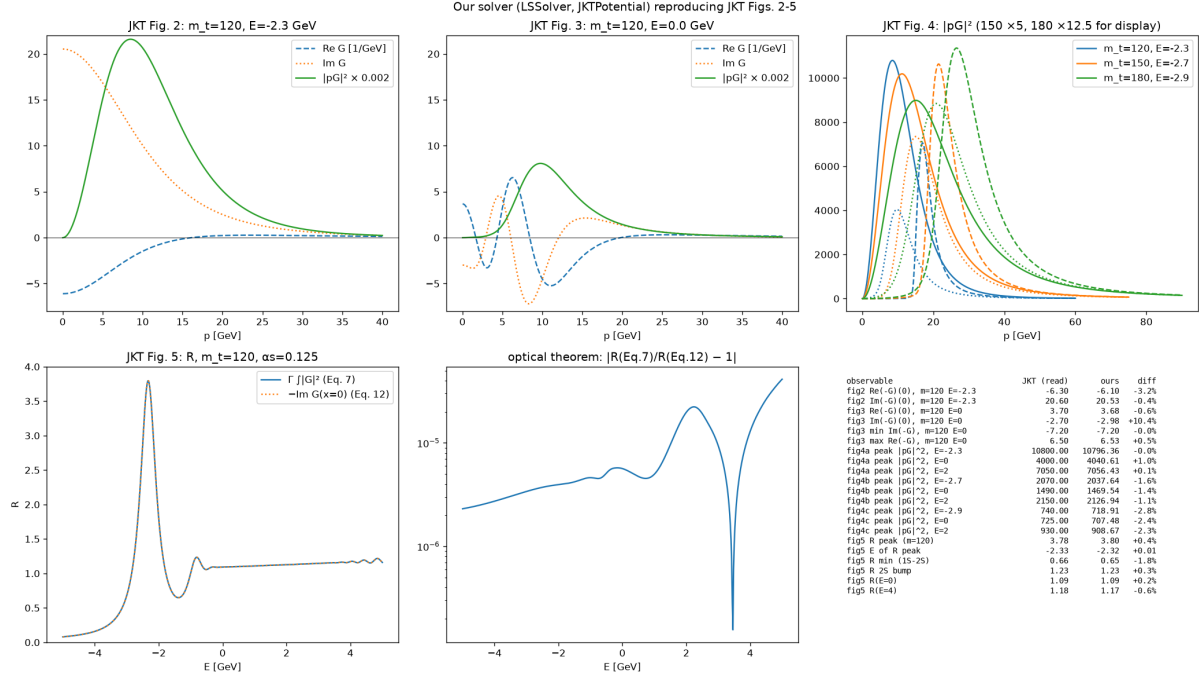


Figure 2: JKT's Figs. 2–5 recomputed with my solver ($\alpha_s = 0.125$, Γ_t from [JKT] (8)). Top: G and $|pG|^2$ at $m_t = 120$ GeV at the 1S peak and at threshold; $|pG|^2$ for $m_t = 120, 150, 180$ GeV (solid, dotted, dashed: below, at and above threshold). Bottom: the cross-section ratio $R(E)$ from $\int \Gamma_t |G|^2$ and from $-\text{Im } G(x=0)$ (the optical theorem [JKT] (12)), their relative difference, and the comparison with values read off JKT's figures. (`1.validate_jkt.py`)

One convention to watch: JKT's Figs. 1–3 plot $-G$ relative to their (6) and (10). At the 1S peak their Fig. 2 has $\text{Im} \approx +20 \text{ GeV}^{-1}$ at $p \rightarrow 0$, whereas $\text{Im } G = -|\psi(0)|^2/\Gamma_t < 0$.

6 Using it for toponium, and the contrast with Fuks et al.

What Fuks et al. do. They re-weight colour-singlet $gg \rightarrow t\bar{t} \rightarrow b\ell^+\nu \bar{b}\ell'^-\bar{\nu}$ MadGraph events by $|G/G_0|^2$ ([Fuks] (15)), with G interpolated from their swData tables by their Fortran routine `CALC GREEN`. They describe the potential behind the tables in four different ways (running tree-level coupling, Richardson following [JKT], fixed $\alpha_s(m_Z) = 0.12$, two-loop) and do not describe the method. I read their tables and run their routine exactly as published, as a reference to compare with. → FuksTables

What is in their tables (Stage 2). Fitting each of their descriptions to swData, using my solver and the exact Coulomb solution (rms of $|\text{swData}/\text{model} - 1|$ on $E \in [-6, 4]$, $p < 30$ GeV):

Potential	rms
fixed- α tree Coulomb, best fit $\alpha = 0.1500$, $m_t = 172.96$, $\Gamma_t = 1.4907$	0.13%
fixed $\alpha = 0.12$ (their Fig. 1 caption), with or without the bracket of (13)	40–50%
JKT potential (13)–(15), $\alpha_s(m_Z) = 0.12$, tree or NLO	55–69%
the same, with a free energy offset	15–17%

So swData is the exact fixed-coupling Coulomb Green’s function at $\alpha = 0.15$, not the JKT potential. A plausible origin: $\alpha_s(m_Z) = 0.12$ run with (14) reaches 0.150 at $Q \approx 25$ GeV, close to the Bohr momentum (Sec. 3). The authors state neither. My solver with Coulomb $\alpha = 0.15$ reproduces their tables as well as the exact solution does (median deviation 2.6×10^{-3} for both, Stage 3, Fig. 3), which is a further check of the solver.

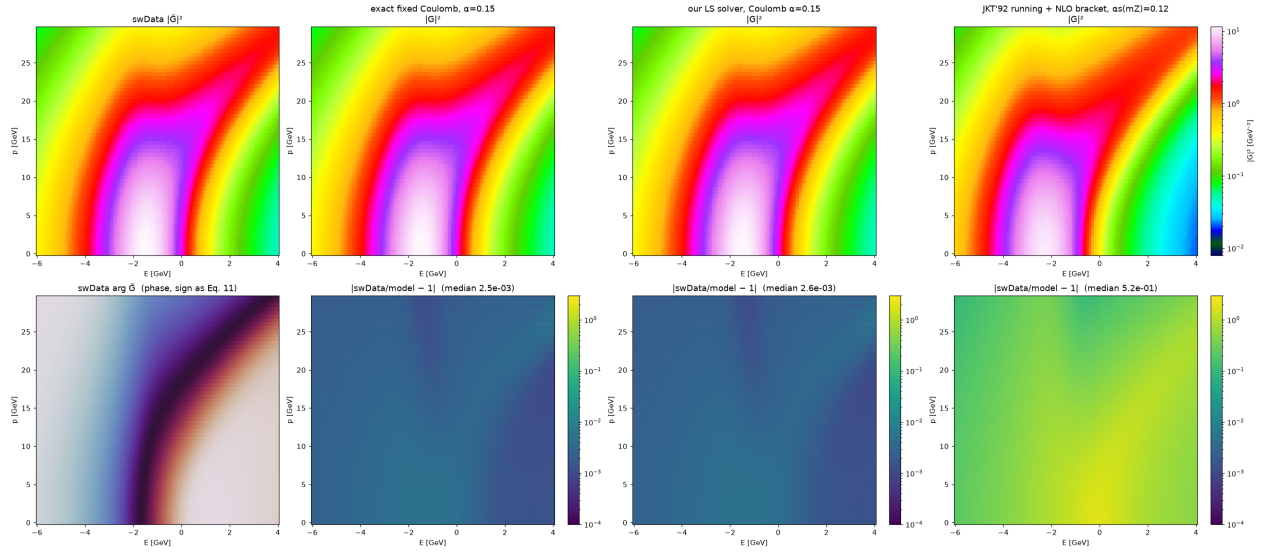


Figure 3: Fuks’s Fig. 1, rebuilt from their swData (left column: $|\tilde{G}|^2$ and phase), next to exact Coulomb at $\alpha = 0.15$, my solver with Coulomb at $\alpha = 0.15$, and my solver with the JKT potential at $\alpha_s(m_Z) = 0.12$. Bottom row: $|\text{swData}/\text{model} - 1|$. (3_fuks_fig1.py)

Events (Stage 4). 13 TeV, CT18NLO, $m_t = 173$, $\Gamma_t = 1.49$: 500k colour-singlet ($330 \leq W \leq 350$ GeV, colour matrix [Fuks] (20)) and 500k full colour ($340 \leq W \leq 350$); the window is imposed by restricting $\tau = \hat{s}/s$ in MadGraph’s `genps.f`. The singlet fraction is 0.281 (expected 2/7). The same events are re-weighted with three sources of G :

	Fuks: swData via CALCGREEN	Coulomb $\alpha = 0.15$ exact	JKT potential (this note) $\alpha_s(m_Z) = 0.12$
$\sigma_{\text{rw}}/\sigma$ ($p^* < 50$ GeV)	8.88	8.90	7.89
$\langle p^* \rangle(E = -2)$ [GeV]	20.7	20.7	19.6
peak of $d\sigma/dW$ [GeV]	344.75	344.75	343.75

With Fuks’s own G the re-weighted distributions reproduce their Figs. 2–5 (Figs. 4–7). The JKT potential, i.e. the physics they cite, gives an enhancement smaller by 11%, a peak 1 GeV lower and softer p^* .

→ Events

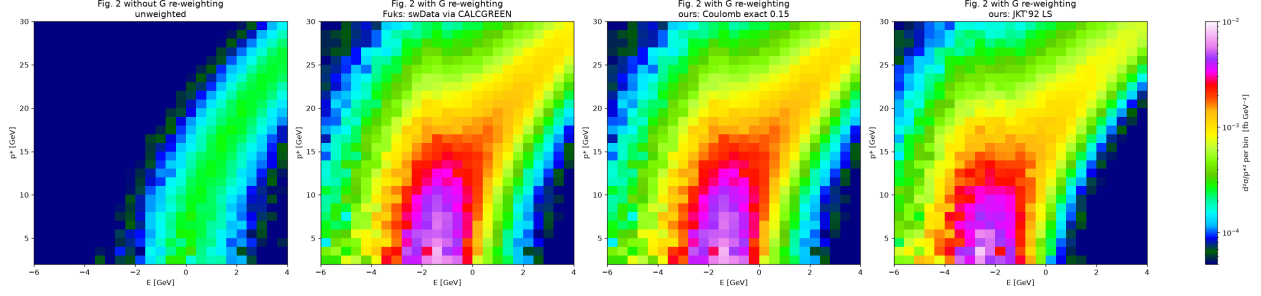


Figure 4: Fuks's Fig. 2 from our events: $d^2\sigma/p^2$ per bin in the (E, p^*) plane, without re-weighting and re-weighted with Fuks's G , exact Coulomb at $\alpha = 0.15$, and the JKT potential. (4.fuks_figs2to5.py)

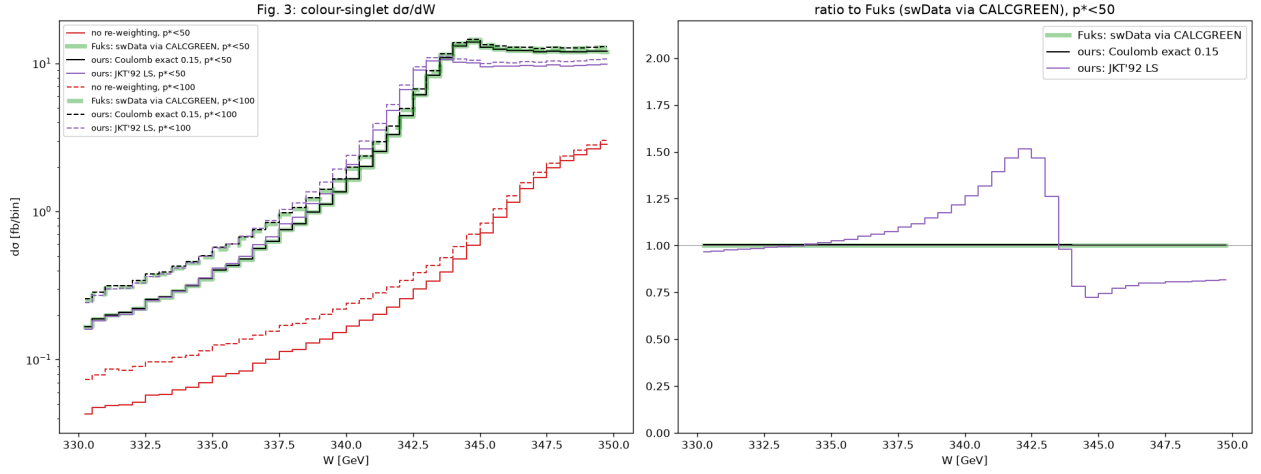


Figure 5: Fuks's Fig. 3: colour-singlet $d\sigma/dW$ in 0.5 GeV bins, without re-weighting and re-weighted with the three sources of G , for $p^* < 50$ and $p^* < 100$ GeV; right, the ratio to Fuks's own G . Exact Coulomb at $\alpha = 0.15$ lies on top of Fuks; the JKT potential moves the peak down by about 1 GeV.

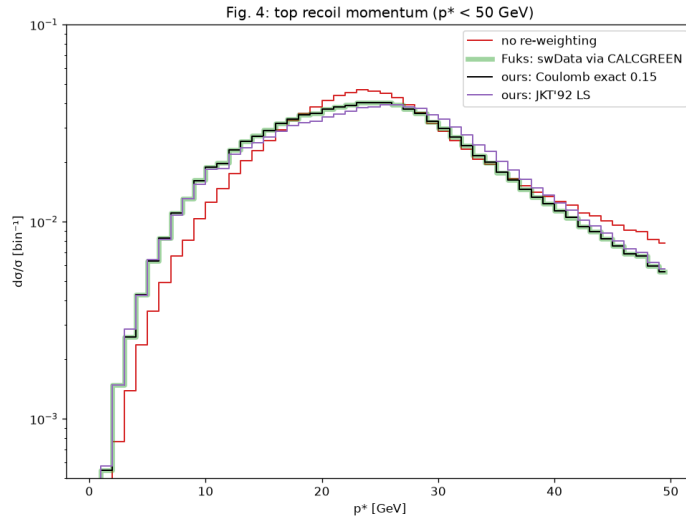


Figure 6: Fuks's Fig. 4: the distribution of the top recoil momentum p^* ($p^* < 50$ GeV, fraction per 1 GeV bin), without re-weighting and re-weighted with the three sources of G .

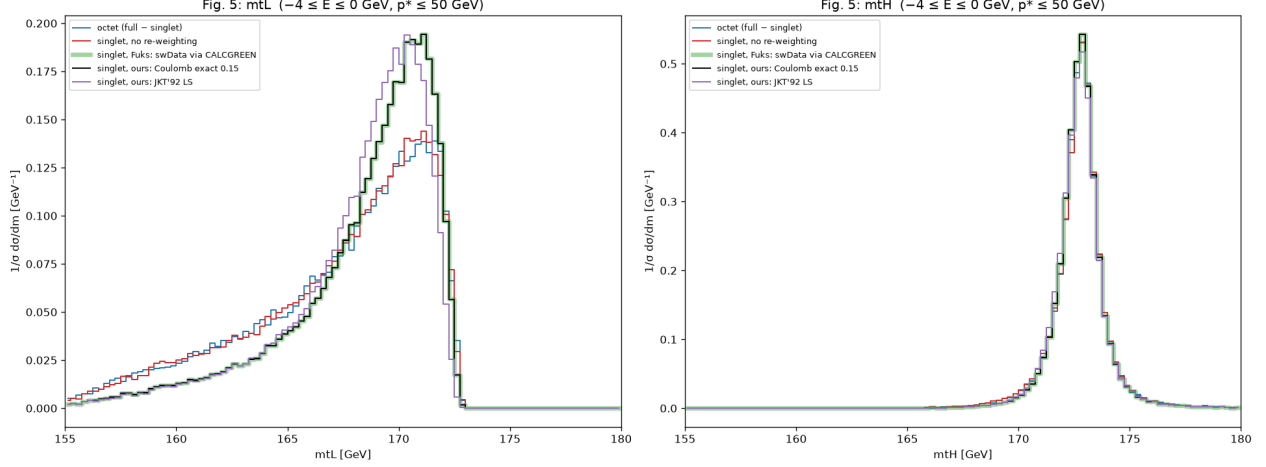


Figure 7: Fuks’s Fig. 5: the lighter and heavier reconstructed top mass, m_{t_L} and m_{t_H} , for $-4 \leq E \leq 0$ GeV and $p^* \leq 50$ GeV: colour octet, singlet without re-weighting, and singlet re-weighted with the three sources of G .

7 Toponium in $pp \rightarrow t\bar{t}$ events: T-factors at 13 TeV

The Green’s function of Secs. 3–4 is put into MadGraph’s matrix element, so that whole event samples contain toponium, and the effect is quoted per bin of measured $t\bar{t}$ distributions as a T -factor,

$$T = \frac{\sigma_{\text{with toponium}}(\text{bin})}{\sigma_{\text{without}}(\text{bin})}.$$

→ 6.tfactors.py

7.1 Colour: which part of $pp \rightarrow t\bar{t}$ can form toponium

At LO, $pp \rightarrow t\bar{t}$ has two channels.

- $q\bar{q} \rightarrow t\bar{t}$: one diagram, $q\bar{q} \rightarrow g^* \rightarrow t\bar{t}$. A single gluon is a colour octet, so the pair it produces is an octet. In the amplitude the colour factor is $(T^a)_{ij}(T^a)_{kl}$; projecting $t_k\bar{t}_l$ on the singlet ($\delta_{kl}/\sqrt{3}$) gives $\text{Tr } T^a = 0$. Singlet $q\bar{q}$ -produced pairs appear only via γ^*/Z (not in our pure-QCD samples) or beyond LO.
- $gg \rightarrow t\bar{t}$: three diagrams, singlet *and* octet ($8 \otimes 8 \ni 1$). MadGraph writes the amplitude in two colour flows, $\mathcal{M} = J_1(T^a T^b)_{kl} + J_2(T^b T^a)_{kl}$, and $|\mathcal{M}|^2 = \sum_{ij} J_i^* C_{ij} J_j$ with

$$C = \frac{1}{3} \begin{pmatrix} 16 & -2 \\ -2 & 16 \end{pmatrix} = \underbrace{\frac{2}{3} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{C_1 \text{ (singlet, [Fuks] (19)–(20))}} + \underbrace{\frac{1}{3} \begin{pmatrix} 14 & -4 \\ -4 & 14 \end{pmatrix}}_{C_8 \text{ (octet)}}, \quad |\mathcal{M}|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_8|^2,$$

with no singlet–octet interference. Near threshold the singlet is about 2/7 of gg .

Only the singlet feels an attractive potential, $V = -C_F \alpha_s / r$, which is the one G is computed for. The octet potential is repulsive, $+\alpha_s / (6r)$: no bound state, a small suppression near threshold, which we (like [Fuks]) neglect. Toponium therefore enters *only* through the gg colour singlet, in the S-wave (1S_0) state that our S-wave G describes.

7.2 The matrix-element modification

Per event, $E = W - 2m_t$ with $W = m_{t\bar{t}}$, and p^* is the top momentum in the $t\bar{t}$ rest frame ([Fuks] (21)). In the window $W \leq 350$ GeV, $p^* \leq 50$ GeV (and $E \geq -20$ GeV, the range of our table) the gg matrix element is changed; outside it, and for $q\bar{q}$, nothing changes:

$$gg: \quad |\mathcal{M}|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_8|^2 \longrightarrow |\mathcal{M}_8|^2 + \left| \frac{G(E, p^*)}{G_0(E, p^*)} \right|^2 |\mathcal{M}_1|^2 = |\mathcal{M}|^2 + \left(\left| \frac{G}{G_0} \right|^2 - 1 \right) |\mathcal{M}_1|^2.$$

$|G/G_0|^2$ is read in the Fortran from a table of our G (bilinear interpolation, a module modelled on [Fuks]’s GREENDATA/CALCGREEN), and $|\mathcal{M}_1|^2$ is computed alongside $|\mathcal{M}|^2$ with C_1 .

Relation to [Fuks] (15)–(16). [Fuks] generate $gg \rightarrow t\bar{t}$ with the singlet colour matrix only. Their (15), $|\mathcal{M}_1|^2 |G/G_0|^2$, is toponium alone; (16), $|\mathcal{M}_1|^2 (|G/G_0|^2 - 1)$, is what must be *added* to an SM sample, the -1 removing the singlet part the SM sample already contains. Our “with” sample is that sum, SM + (16), built into one matrix element: the octet term above is the octet of their SM sample, not extra physics, and the double counting is removed in the singlet only, as in [Fuks]. Building it in keeps $|\mathcal{M}|^2 \geq 0$ ((16) alone is negative where $|G/G_0| < 1$, which MadGraph cannot generate). Other differences: [Fuks] use decay chains (W near on-shell) and $\ell = e, \mu$; we generate $t\bar{t} \rightarrow be^+ \nu \bar{b} \mu^- \bar{\nu}$ with the full W propagators, and the branching ratio cancels in T .

7.3 α_s : CTEQ’s running versus [JKT] (14)

0.118 is CT25NNLO’s $\alpha_s(m_Z)$; its $\alpha_s(Q)$ runs, but differently from [JKT] (14): CTEQ solve the renormalisation-group equation numerically at three loops with a variable number of flavours (thresholds at m_c, m_b) and tabulate it; (14) is a two-loop closed form with fixed $n_F = 5$. Both agree at m_Z by construction:

Q [GeV]	5	10	25	91.19
CT25NNLO (LHAPDF)	0.21257	0.17815	0.14688	0.11800
[JKT] (14), two-loop, $\alpha_s(m_Z) = 0.118$	0.21396	0.17864	0.14705	0.11800

MadGraph takes $\alpha_s(Q)$ for the matrix elements from the PDF. For one α_s everywhere, the potential (13) behind G uses CT25’s $\alpha_s(Q)$ too, read from LHAPDF (it enters only above 5 GeV and through C). This changes C from -0.0320 to -0.0332 and $|G/G_0|^2$ by at most 0.1%. Formally it combines three-loop running with the one-loop bracket of (13); the effect is far below the other approximations.

→ JKTPotential(alpha_fn=...)

7.4 From events to T-factors

1. *Generate.* Two samples of the same process, “without” and “with” toponium, 10 MadGraph runs of 10^6 events each, every run with its own seed; LO, CT25NNLO, $m_t = 173$, $\Gamma_t = 1.49$ GeV, no generation cuts. The toponium effect is entirely in *which* events MadGraph generates (it unweights with the modified $|\mathcal{M}|^2$); no reweighting is done afterwards.
2. *Reconstruct.* From each event’s final state: top = $b + e^+ + \nu_e$ (PDG ids 5, $-11, 12$), antitop = $\bar{b} + \mu^- + \bar{\nu}_\mu$ ($-5, 13, -14$). The tops are off-shell: $m(be^+ \nu)$ has median 173.04 GeV and 90% of events within 169–179 GeV.
3. *Observables.* From the summed four-momenta: $m_{t\bar{t}}, y_{t\bar{t}}, |y_{t\bar{t}}|, |y_t + y_{\bar{t}}|/2, H_T = p_T^t + p_T^{\bar{t}}, p_T^t, |y_t|$ and the 2D $|y_t| \times p_T^t$, in all the requested binnings (12).

4. *Count.* Every event weight equals the run’s cross section; dividing by the total number of events,

$$\sigma_{\text{bin}} = \sum_{i \in \text{bin}} w_i \text{ and its error is } \sqrt{\sum w_i^2}.$$

5. *Divide.* $T = \sigma_{\text{with}}/\sigma_{\text{without}}$ per bin; the bin widths cancel.

7.5 Results

Three runs at 13 TeV, 10M + 10M events each: run 1 is $pp \rightarrow t\bar{t}$ with full colour (gg singlet and octet, and $q\bar{q}$); run 2 is $gg \rightarrow t\bar{t}$ in the colour singlet only (no octet, no $q\bar{q}$); run 3 is run 1 with the octet threshold effect added (Sec. 7.6), sharing run 1’s “without” sample. In runs 1 and 2 the toponium excess $\Delta\sigma$ is the same, as it must be, since only the gg singlet is changed; the denominators differ. Run 3 adds the (negative) octet change.

	run 1: pp , singlet	run 2: gg , singlet only	run 3: pp , singlet + octet
$\sigma_{\text{without}}, \sigma_{\text{with}}$ [pb]	5.1229, 5.1702	0.9903, 1.0366	5.1229, 5.1633
$\Delta\sigma$ [pb]	0.0473	0.0463	0.0404
total $T - 1$	+0.92%	+4.68%	+0.79%
$m_{t\bar{t}} \in [250, 400]$ GeV	$+4.57 \pm 0.10\%$	$+20.9 \pm 0.1\%$	$+3.87 \pm 0.10\%$
$m_{t\bar{t}} \in [345, 400]$ GeV	$+2.47 \pm 0.10\%$	$+11.2 \pm 0.1\%$	$+1.91 \pm 0.10\%$
$H_T \in [0, 90]$ GeV	$+7.0 \pm 0.13\%$	$+36.9 \pm 0.2\%$	$+6.0 \pm 0.13\%$
$p_T^t \in [0, 60]$ GeV	$+4.26 \pm 0.10\%$	$+22.2 \pm 0.1\%$	$+3.63 \pm 0.10\%$
$ y_{t\bar{t}} $ bins	$\approx +0.9\%$, nearly flat	$+4.2\%$ to $+6.8\%$	$\approx +0.75\%$, nearly flat

The effect sits where the tops are slow: the lowest $m_{t\bar{t}}$ bin (which contains the whole window) and the lowest H_T bin; higher $m_{t\bar{t}}$ and H_T bins are unchanged within statistics. In rapidity it is spread over all bins, rising towards large $|y|$ because the low-mass toponium pairs reach larger rapidities. The $m_{t\bar{t}}$ binning that starts at 345 GeV shows about half the effect of the one starting at 250 or 325 GeV: most of the toponium peak ($W \approx 343\text{--}345$ GeV) lies below its first edge. For a pp measurement, run 3’s T-factors (singlet and octet) are the most complete, run 1’s are those of [Fuks]’s convention (singlet only); run 2 isolates the channel in which toponium forms. Per-bin values: `results/6_tfactors/<run>/tfactors_13TeV.csv`.

7.6 The colour-octet threshold effect

The octet (gg octet and all of $q\bar{q}$) also has a Coulomb interaction near threshold, but a *repulsive* one, $V_8 = +\alpha_s/(6r)$ (colour factor $C_F - C_A/2 = -1/6$ instead of $C_F = 4/3$): no bound states, and a suppression of the rate near threshold. [Fuks], and runs 1–2, neglect it. To measure it, run 3 repeats run 1’s “with” sample (10M events, seeds 6301–6310) with, in the same window, both colour states changed:

$$gg : |\mathcal{M}|^2 \rightarrow \left| \frac{G_1}{G_0} \right|^2 |\mathcal{M}_1|^2 + \left| \frac{G_8}{G_0} \right|^2 |\mathcal{M}_8|^2, \quad q\bar{q} : |\mathcal{M}|^2 \rightarrow \left| \frac{G_8}{G_0} \right|^2 |\mathcal{M}|^2.$$

G_8 is the exact Coulomb solution (Sec. 5.1) for the repulsive potential, with $\alpha_s = 0.147$, CT25NNLO’s α_s at 25 GeV (the Bohr scale); it agrees with our solver to 2×10^{-6} and gives $|G_8/G_0|^2 \approx 0.67\text{--}0.82$ in the window. The denominator is run 1’s “without” sample, so $\Delta T = T_{\text{run 3}} - T_{\text{run 1}}$ isolates the octet. (A check with $|G_8/G_0|^2$ set to 10 raised σ by the predicted +0.26 pb, so the octet code acts on both gg and $q\bar{q}$.)



Figure 8: Run 1 ($pp \rightarrow t\bar{t}$, full colour): $T = \sigma_{\text{with}}/\sigma_{\text{without}}$ per bin for all requested binnings (the 2D $|y_t| \times p_T^t$ binning in slices of $|y_t|$); error bars are MC statistics of the two independent samples.



Each bar: $T = \sigma_{\text{with toponium}} / \sigma_{\text{without}}$ in that bin ($T = 1$: no change).
Orange: toponium adds events; blue: fewer.
Error bars: Monte Carlo statistics.
 $gg \rightarrow t\bar{t}$, colour singlet only;
1.3 TeV, LO MadGraph, CT2SMNLO with its alpha_s(Q);
10M events without + 10M with toponium.

Figure 9: Run 2 ($gg \rightarrow t\bar{t}$, colour singlet only): T per bin for the same binnings. The toponium excess is the same as in run 1; the smaller denominator makes the relative effect about five times larger.

	singlet only (run 1)	singlet + octet (run 3)	ΔT
total	1.00923	1.00788	-0.0014
$m_{t\bar{t}} \in [250, 400]$ GeV	1.0457	1.0387	-0.0070 ± 0.0010
$H_T \in [0, 90]$ GeV	1.0702	1.0601	-0.0101 ± 0.0013
$p_T^t \in [0, 60]$ GeV	1.0426	1.0363	-0.0063 ± 0.0010
rapidity bins	≈ 1.009	$\approx 1.007\text{--}1.008$	≈ -0.001 to -0.002

The octet changes the cross section by $\Delta\sigma_8 = -0.0069 \pm 0.0008$ pb, i.e. -14.6% of the singlet toponium excess (0.0473 pb); rescaling with $|G_8/G_0|^2$ on the events in the window gives -12% ($\alpha_s = 0.12$) to -17% ($\alpha_s = 0.18$). About 84% of the threshold-window rate is octet. So: in absolute terms the octet changes every T-factor by less than 1% (at most -1.0% , in the lowest H_T bin), but relative to the toponium signal it is a $\sim 15\%$ effect, and it always *reduces* the signal. Neglecting it is justified only at the level of a $\sim 15\%$ uncertainty on $T-1$; including it (run 3) is the more complete description. Comparison plots: `results/6_tfactors/run3_pp_singlet_octet.13TeV/plots/compare_*.png` (Fig. 10); run 3’s own T-factors: Fig. 11.

Caveats.

1. LO matrix elements. T is a ratio meant to multiply higher-order $t\bar{t}$ predictions, assuming higher-order corrections largely cancel between numerator and denominator.
2. The window is a hard cut ($W \leq 350$ GeV, $p^* \leq 50$ GeV): the factor is switched off, not faded to 1, outside it.
3. Runs 1–2 neglect the octet’s repulsive Coulomb interaction (as [Fuks] do); run 3 measures it: -15% of the toponium effect, below 1% in every T (Sec. 7.6).
4. The two samples are independent: in bins where nothing changes, T scatters around 1 by its statistical error (a few % in the sparse high- $m_{t\bar{t}}$ bins), although in bins lying entirely above $m_{t\bar{t}} = 350$ GeV the true value is exactly $T = 1$, and should be quoted as such.

8 Where I fill in or depart from JKT

1. α_s is specified at m_Z and $\Lambda_{\overline{\text{MS}}}$ obtained by solving (14) there; JKT quote α_s directly.
2. Grid: JKT’s subintervals split once more, at 4 and 60 GeV (Step 5).
3. $\mathcal{A}(p)$ on a graded auxiliary Gauss–Legendre rule (Step 4); JKT only say “numerical integration”.
4. K between the nodes by cubic-spline interpolation (Step 7); JKT do not say how they evaluate off the grid.
5. In the T-factor runs (Sec. 7) only, $\alpha_s(Q)$ in (13) is CT25NNLO’s (LHAPDF), not (14). Everything else (the potential, the cut and its normalisation, the subtraction, the linear system) is [JKT] as printed.

A The code, file by file

Everything lives flat in `toponium_study/`. `toponium.py` is the only library; `0...6_*.py` are the seven scripts you run (`uv run python 0_solver_checks.py`, and so on), each writing `results/<stage>/summary.txt` and figures. Every class opens with an “EQUATION MAP” listing which paper equation each method implements.



Figure 10: T per bin with toponium in the colour singlet only (run 1, orange) and with the singlet and the repulsive octet (run 3, purple), for all binnings; same “without” sample, error bars are MC statistics. The octet lowers T only where toponium acts (lowest $m_{t\bar{t}}$, p_T^t , H_T bins, and slightly in rapidity), by $\sim 15\%$ of $T - 1$.



Figure 11: Run 3 ($pp \rightarrow t\bar{t}$, singlet $\times |G_1/G_0|^2$ and octet $\times |G_8/G_0|^2$): T per bin for all binnings.

toponium.py: six classes, top to bottom

Constants

m_t , Γ_t , C_F , n_F , and G_0 , [JKT] (6).

JKTPotential

Sec. 3: `alpha_2loop`, `lambda_msbar_2loop` (14); `V_pert` (13); `V_ir`, `continuity_constant_C`, `V_jkt` (15); `V_cut` (19); `V_cut_position`, `energy_shift` (20). `JKTPotential.coulomb(alpha)` gives fixed-coupling Coulomb with the same interface, to test the solver.

LSSolver

Sec. 4: `kernel` (18), `jkt_grid` (23), `script_A` (22), `solve_K` (21), `G` (16). Usage: `LSSolver(potential).G(E, p)`.

CoulombExact

Sec. 5.1: the exact Coulomb G (`G`, fast; `G_adaptive`, slow reference).

FuksTables

Fuks's `swData` and `CALCGREEN`, used as-is (reference only; nothing of mine is in it).

Events

the MadGraph events, read from `results/4_fuks_figs2to5/_cache*.npz`, and the weight $|G/G_0|^2$ for G from any of the above.

The seven stage scripts

script	what it does
<code>0.solver_checks.py</code>	solver vs exact Coulomb, convergence, q_{cut} independence (Sec. 5); <code>kernel_subtraction.png</code>
<code>1.validate_jkt.py</code>	solver vs JKT's own Figs. 2–5; <code>jkt_figs.png</code>
<code>2.identify_swdata.py</code>	which potential is in Fuks's tables (Sec. 6)
<code>3.fuks_fig1.py</code>	Fuks's Fig. 1 rebuilt from <code>swData</code> , my G alongside
<code>4.fuks_figs2to5.py</code>	events re-weighted with the three sources of G . Reads <code>_cache_full.npz</code> , <code>_cache_singlet330.npz</code> : the only copy of the events, never delete them
<code>5.comparison_tables.py</code>	CSV tables for comparing with an independent solver (colour singlet): Λ , C , C_1 , $\tilde{V}_0$ per setting; $\alpha_s(Q^2)$ and each piece of the potential vs Q^2 ; G on (E, p) grids
<code>6.tfactors.py</code>	toponium T-factors at 13 TeV: LO MadGraph $pp \rightarrow t\bar{t} \rightarrow be^+\nu\bar{b}\mu^-\bar{\nu}$ without, and with the gg matrix element changed to $ M ^2 + (G/G_0 ^2 - 1) M_1 ^2$ ([Fuks] (16), singlet (19)–(20)). Here (13) uses CT25NNLO's own $\alpha_s(Q)$ (from LHAPDF) instead of (14), the same α_s MadGraph uses. Runs: <code>run1_pp_fullcolour_13TeV</code> , <code>run2_gg_singlet_13TeV</code> , <code>run3_pp_singlet_octet_13TeV</code>

`reference/` holds both PDFs and `swData`; `README.md` has the recipe to regenerate the MadGraph samples; `pyproject.toml/uv.lock` define the Python 3.11 environment.